

After completing the preparation materials for the Pharmacology of Antiseizure Drugs, students should be able to:

1. Explain the various mechanisms of antiseizure drug action at the levels of specific ion channels and neurotransmitter systems.
2. Describe the main pharmacokinetic features of the antiepileptic drugs and their metabolic- and transport-related drug interactions.
3. Compare and contrast the drugs used in the treatment of focal onset seizures, generalized onset seizures, various epileptic syndromes, and status epilepticus by giving specific examples.
4. Correlate the adverse effects of the drugs covered in this lecture (the “starred drugs”) with their mechanisms of action
5. Explain the meaning and clinical relevance of HLA genetic polymorphisms on adverse event risk.
6. Apply the distinctive class and drug-specific toxicities to the selection of antiepileptic therapy for the individual patients, including children, adults, pregnant patients, and older patients.

Pharmacokinetics of Valproate

- Oral; I.V.
- Protein binding 80-92% (saturable, concentration-dependent)
- Hepatic metabolism:
 - mitochondrial β -oxidation
 - glucuronide conjugation
 - other pathways via several CYPs to a lesser extent
- Bioavailability ~90%
- $t_{1/2}$ varies:
 - adults 9-16h
 - children 7-13h
 - prolonged in neonates and patients with liver disease

Main metabolic pathways

What is the take-away? Monitor blood levels of the drug.

Valproic acid:

Multiple cellular mechanisms observed in model systems

1. VG Na⁺ channel block suppresses high-frequency, repetitive neuronal firing (different sites than phenytoin and carbamazepine)
2. Reductions in voltage-gated T-type Ca²⁺ currents
3. Increases GABA concentrations, stimulates GABA synthesis, and inhibits GABA degradation

Other potential actions observed in experimental systems.

The relevance of these actions to antiseizure activity is not established.

Valproate is considered
first-line for treating a range of seizure types in adults and children

Monotherapy or adjunctive therapy in:

- Focal-onset seizures
- Generalized-onset tonic-clonic seizures
- Absence seizures, simple or complex
 - Valproic acid and ethosuximide are first-line agents
- Myoclonic epilepsy syndromes / Atonic seizures
- Status epilepticus (valproate I.V., off-label alternative)

Non-seizure applications:

- Acute mania associated with bipolar disorder
- Migraine acute treatment and prophylaxis

Takeaway:

Valproate is useful for many seizure types and it has non-seizure applications.

Valproate Adverse Effects (not a complete list)

- Common:
 - nausea, vomiting, abdominal pain and heartburn (gradually titrate dose)
 - sedation, dizziness, ataxia, cognitive impairment
- Tremor, hair loss, and weight gain (4-9% of patients)
- Hepatotoxicity (range: from an ↑ in LFTs to fulminant liver failure and death)
- Hematological: Thrombocytopenia / Reduced platelet function
- Hyperammonemia / encephalopathy (related to inhibition of mitochondrial carbamoyl phosphate synthase necessary for ammonia elimination; dose-dependent effect)
- Acute pancreatitis, rare (US Boxed Warning)
- Hypersensitivity reactions: Rash, SJS, TEN, DRESS
- Caution: Patients with hereditary neurometabolic syndromes caused by mitochondrial DNA polymerase gamma (POLG) gene mutations are at increased risk of acute liver failure and death – valproate is contraindicated

Valproate: additional cautions

PREGNANCY: *Teratogenic*

Neural tube defects (spina bifida)

Craniofacial defects (eg, cleft palate)

Cardiovascular malformations

Other body malformations

Lower cognitive test scores

Neurodevelopmental defects

DRUG INTERACTIONS:

Valproate inhibits the metabolism of many drugs, including lamotrigine, phenobarbital, and ethosuximide → may increase the serum levels of the drugs.

Lamotrigine:

- Practice point: Slow titration of lamotrigine dose when added to valproate therapy

Pharmacokinetics

Favorable PK profile:

- good absorption
- good bioavailability
- low plasma protein binding
- minor non-CYP hepatic metabolism
- renal excretion ~70% unchanged
- may undergo proximal tubule reabsorption → $t_{1/2}$ 20-30h

Mechanisms

Antiseizure effects are likely due to a combination of actions

- VG Na⁺ channel inhibition
- Depresses glutamate AMPA/kainite excitatory action
- Enhances GABA_A inhibitory effect
- Carbonic anhydrase inhibition (weak) – antiseizure relevance unknown

Topiramate Therapeutic uses in adults and children

Seizure applications

- Focal-onset seizures
- Generalized-onset tonic-clonic seizures
- Lennox-Gastaut syndrome (adjunctive therapy)
- May be effective in other childhood seizure types

Non-seizure applications:

- Prophylaxis of migraine and cluster headaches
- Weight loss
(in fixed-combination with phenteramine, a sympathomimetic similar to amphetamine)
- Binge eating disorder
- Alcohol use disorder
- and others

Topiramate: Pregnancy and Drug Interactions

Pregnancy	Adverse effects in animal reproduction studies • May cause fetal harm: oral clefts reported • Consider alternative medication for patients who wish to become pregnant. • Effective contraception recommended for people who or reproductive potential who are not planning to become pregnant. • Pregnant women and newborns should be monitored for metabolic acidosis, which may be induced by topiramate.
Drug Interactions	• Enzyme inducers increase clearance • Concurrent use of topiramate and valproic acid may increase the risk of hyperammonemia and associated encephalopathy. • CNS depressant effects of other drugs may enhance topiramate's depressant effects.

Questions

Thinking of valproate's mechanisms, therapeutic uses, and side effects, what are three takeaways regarding valproate?

Thinking of topiramate's mechanisms, therapeutic uses, and side effects, what are three takeaways regarding topiramate?

What about usage in pregnancy or individuals who may become pregnant?

Felbamate pharmacology

PK:

excellent bioavailability
CYP3A4 metabolism
20-hr half-life (adults)

MOA:

1. Use-dependent block of NMDA glutamate receptor
2. Potentiation of GABA_A-mediated inhibitory effect

Therapeutic uses:

Seizure disorder refractory to other ASDs

🔔 Informed consent must be obtained

- Focal-onset seizures with or without generalization
- Seizures of Lennox-Gestaut syndrome

High rates of fatal aplastic anemia and hepatitis failure compared with those treated with other ASDs.

- GI / CNS side effects
- DDIs:
CYP3A4, CNS depressants

Gabapentin / Pregabalin

PK: Uptake via L-amino acid transporter (saturable)

Not metabolized

Renal excretion 100% unchanged

Nonseizure: Neuropathic pain and several other off-label uses

MOA: Block $\alpha 2\delta$ subunit on VG Ca^{2+} channel on presynaptic neuron $\rightarrow$ may $\downarrow$ synaptic release of excitatory neurotransmitters

May involve other mechanisms.

USES:

Focal-onset seizures (adjuncts)

Nonseizure: Neuropathic pain, restless legs syndrome, hot flashes of menopause, and many other uses

ADVERSE EFFECTS

- Sedation; dizziness; weight gain; peripheral edema

FDA warning: Respiratory depression; increased risk with opioids, CNS depressants; elderly

PREGNANCY

- Adverse events observed in animal reproduction studies
- Pregnancy registry outcome data is limited

DRUG INTERACTIONS

- No known metabolic interactions

The precise mechanism for the gabapentinoids' actions is not known. 23

Levetiracetam pharmacology

Drug binding to SV2A appears to reduce the synaptic release of glutamate and GABA.

PK:

- oral, 100% bioavailability
- I.V.
- nonhepatic hydrolysis
- renal, unchanged drug
- $t_{1/2}$ ~7h (2x daily dosing)

- TU: broad spectrum

Adjunctive use

- Focal-onset seizures
- Generalized-onset tonic-clonic seizures
- Myoclonic seizures
- AE: Mild-moderate, often subside
 - Fatigue, irritability, somnolence, dizziness
 - Psychotic symptoms
- DDI: Low potential for metabolic drug interactions

Ethosuximide pharmacology

Pharmacokinetics	Oral CYP3A4 metabolism Renal excretion of metabolites (small amount unchanged) $t_{1/2}$ 50-60 h (adults), 30 h children
Mechanism of action	See next slide
Therapeutic use	Absence seizures
Adverse effects	Most common: Nausea and vomiting somnolence, sleep disturbance, hyperactivity
Drug interactions	CYP3A4-mediated Ethosuximide does not induce or inhibit metabolic enzymes

➤ Practice point: Ethosuximide is often given 2 to 3 times daily to reduce GI side effects.

Questions

A particular broad-spectrum ASD that inhibits NMDA and potentiates GABA at GABA_A receptors is used only in refractory seizures, such as Lennox-Gastaut syndrome. Informed consent is required. Why?

A particular class of drugs that acts at a presynaptic site is used for neuropathic pain, a number of other nonseizure uses, and as add-on therapy for management of focal seizures. What is its mechanistic target?

A broad spectrum ASD with 100% bioavailability, nonhepatic metabolism, renal excretion of unchanged drug has mild-moderate side effects and low potential for drug interactions. What are the drug and the mechanistic target?

What are the most common side effects of the narrow spectrum drug for absence seizures?

Pharmacokinetics: Phenobarbital and Primidone

	Phenobarbital	Primidone
A	Oral (good bioavailability); I.V.; I.M. I.V. onset of action ~5 minutes	Oral only, 90-100% bioavailability
D	Protein binding ~50% bound V_d is larger in neonates and infants than in older individuals	Protein binding ~30%
M	Hepatic via CYP2C9	Hepatic to phenobarbital and PEMA* All 3 compounds are active
	Phenobarbital and primidone (and all other barbiturates) are strong inducers of CYP1A2, 2B6, 2C8/9, 3A4, UGT, and P-glycoprotein	
E	Renal 25-50% unchanged	Renal unchanged drug, phenobarbital, and PEMA*
$t_{1/2}$	1½-5 days in adults Less in infants and children	Primidone 5-15 hours; PEMA 16 hours Age-dependent

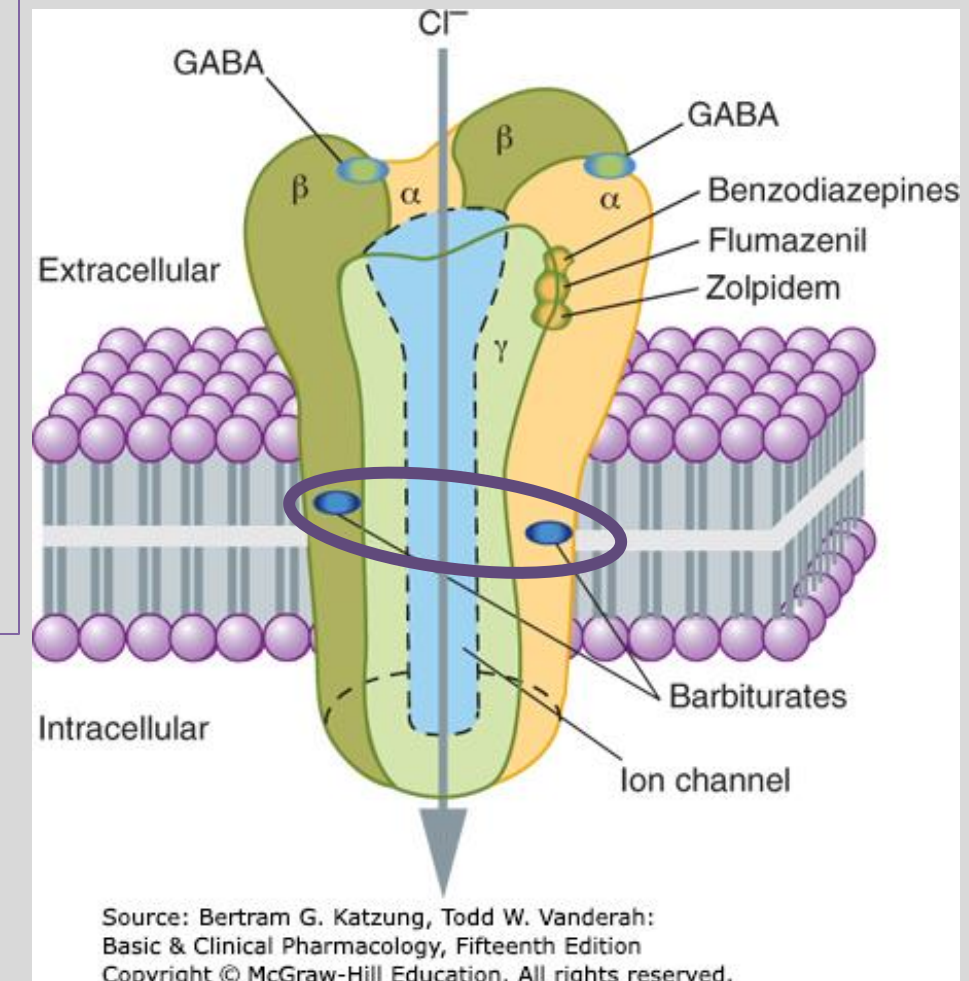
*PEMA: phenylethylmalonamide

Phenobarbital: GABA_A receptor positive allosteric modulator

- Barbiturates bind to allosteric site on both α and β subunits;
- prolong duration*** of the channel openings permitting Cl⁻ influx and hyperpolarization → enhanced GABA-mediated inhibition
- High-doses: Barbiturates also have GABA-mimetic activity (direct stimulation of GABA_A receptors)

Therapeutic uses:

- Focal-onset seizures
- Generalized tonic-clonic seizures (especially children)
- Status epilepticus (alternative)



Katzung Figure 22-6: A model of the GABA_A receptor-chloride ion channel macromolecular complex

Phenobarbital Adverse Effects

- Sedation
- Depression of mood
- Cognitive impairment
- Hyperactivity particularly in children and patients in acute pain
- Agitation, confusion in elderly
- Porphyrias (barbiturates can precipitate symptoms)

- Rash
- Hepatotoxicity
- Bone marrow toxicity

Intravenous

- **Respiratory depression** – esp when used after benzodiazepine
- **Hypotension**

POTENT INDUCER of CYPs
Positive evidence of human fetal risk

Pharmacokinetics of Benzodiazepines

	Diazepam	Lorazepam	Midazolam
A	IV, IM, rectal, intranasal, oral	IV, IM, oral, sublingual	IV, IM, intranasal, buccal
Onset	Immediate (I.V.)	5-10 min (I.V.)	IV 3-5 min, IM ~15 min
Duration of antiseizure effect	15-30 min	6-8 hours	IV <2 h, IM up to 6 h
M	CYP2C19, 3A4, active metabolite, glucuronidation	glucuronide conjugation	CYP3A4, active metabolites
E	Urine, glucuronide metabolites	Urine, glucuronide metabolites	Urine, metabolites
t _{1/2}	Diazepam: ~40 hours Nordiazepam ≥3 days	~20 hours (adults)	2-7 h

Note: Diazepam accumulates with multiple dosing and the $t_{1/2\text{terminal}}$ is slightly prolonged as the drug moves out of fat stores.

Diazepam: Rapid Termination of Effect by Redistribution

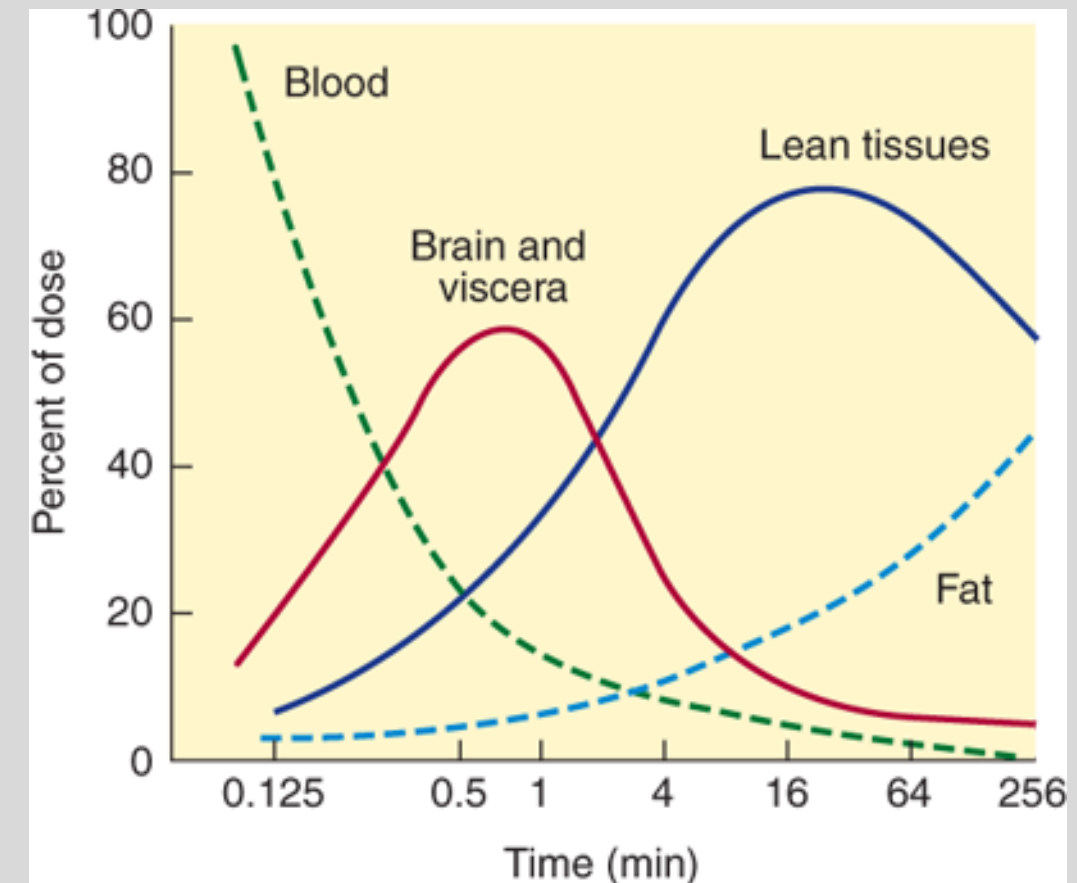
1) Diazepam is very lipid soluble so the initial rapid distribution of a single bolus dose into highly perfused CNS tissue produces an immediate effect.

- Then,

2) As plasma levels of drug decrease, diazepam moves down its concentration gradient out of the brain tissue terminating the effect of the single bolus dose.

- Diazepam rapidly redistributes into the less perfused muscle and fat tissues.

A single bolus dose of diazepam has a short duration of anti-seizure effect in status epilepticus, about 15-30 minutes, despite its long half-life.



Source: Todd W. Vanderah:
Basic & Clinical Pharmacology, Sixteenth Edition
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Benzodiazepines therapeutic uses

Status epilepticus:

Bolus dosing of benzodiazepines for initial treatment

Diazepam:

- Immediate onset
- Short duration of action (15-30 min) is a disadvantage

Lorazepam:

longer duration of antiseizure action (6-8 h)

Midazolam:

IM, intranasal, buccal alternative if IV access not available

rapid onset; short $t_{1/2\text{elimination}}$ 2-7h

Chronic epilepsy management

Oral therapy

Clonazepam and Clorazepate:

Lennox-Gastaut syndrome

Atonic and myoclonic seizures

Monotherapy or adjuncts with other ASDs

Clobazam:

Lennox-Gastaut syndrome

Treatment-refractory seizures

Adjunctive use

Nonseizure applications

Anxiety; panic disorder; alcohol withdrawal; muscle relaxant (esp diazepam); sedative-hypnotic; preop sedation, anxiolysis, and amnesia; induction of anesthesia (balanced anesthesia); sedation for mechanically ventilated patients

Benzodiazepines side effects and cautions

Oral / rectal:

- Sedation
- Depression
- Incoordination
- Behavioral disturbances in children

Clobazam:

- Severe skin reactions

Parenteral:

Acute hypotension

Ataxia

Respiratory depression

due to rapid rise in CNS concentrations

➤ *Gradually titrate to effect to avoid respiratory depression.*

Caution:

Benzodiazepines may produce withdrawal seizures upon discontinuation of the drug.

Drug interactions

Drugs with similar effects

Drugs that induce or inhibit hepatic enzymes

PREGNANCY

- Teratogenic effects observed (additional studies are needed)
- Hypoglycemia, respiratory problems
- Neonatal withdrawal symptoms
- Hypotonia: Floppy baby syndrome

Withdrawal symptoms may occur days to weeks after birth.



Cannabidiol (pharmaceutical) pharmacology

PK: Oral, high protein binding, extensive distribution and hepatic metabolism CYP3A4, active metabolite, feces, $t_{1/2}$ ~60 h

MOA: Unknown
May act on several endogenous systems
(It does not appear to directly activate CB1 or CB2 receptors.)

Therapeutic uses:

- Seizures associated with Dravet syndrome, Lennox-Gestaut syndrome, or tuberous sclerosis complex in patients ≥ 1 year of age.

- Sedation, lethargy
 - Elevated liver transaminases
- DDIs: high potential
- Valproate: enhanced hepatotoxic effects
 - CYP2C19: clobazam, citalopram, clopidogrel (antiplatelet drug)
 - CYP3A4: numerous
 - CNS depressants

Questions

What is the mechanism of action of GABA_A receptor positive allosteric modulators?

What drugs fall into this class?

What are the first-line drugs for the treatment of status epilepticus?

When treating status epilepticus, what drug-drug combination increases the risk of respiratory depression?

What is the explanation for the short duration of a single bolus dose of diazepam?

What are the antiseizure indications for pharmaceutical cannabidiol?

Drugs Effective for Seizures

Drugs for Focal-Onset Seizures

Carbamazepine	Phenytoin
Oxazepine	Gabapentinoids
Eslicarbazine	Cenobamate

Drugs for Focal-Onset Seizures and Some Generalized-Onset Seizures

Lamotrigine	Phenobarbital
Lacosamide	Primidone
Levetiracetam	Felbamate
	Perampanel

Benzodiazepines

Diazepam	Clobazam
Lorazepam	Clonazepam
Midazolam	Clorazepate

Drugs for Generalized-Onset Seizures

Valproate
Topiramate
Zonisamide

Drugs for Generalized Absence Seizures

Ethosuximide
Valproate

Drugs for Other Types of Seizures

Cannabidiol (pharmaceutical)	Rufinamide
Clobazam	Stiripentol
Benzodiazepines	Fenfluramine

Summary of Adverse Effects

Antiepileptic Drug	Adverse Effects
Benzodiazepines	Sedation, tolerance, dependence
Carbamazepine	Diplopia, cognitive dysfunction, ataxia; rare occurrence of severe blood dyscrasias and SJS/TEN/DRESS syndrome; hyponatremia; induces hepatic drug metabolism
Ethosuximide	Nausea, lethargy, headache, dizziness
Felbamate	Aplastic anemia, hepatic failure
Gabapentin	Dizziness, sedation, ataxia, nystagmus; does not affect drug metabolism (pregabalin is similar)

Summary of Adverse Effects of Commonly Used ASDs

Antiepileptic Drug	Adverse Effects
Lamotrigine	Dizziness, ataxia, nausea, rash, rarely SJS/TEN; slow titration of dose to reduce risk of rash
Levetiracetam	Dizziness, sedation, weakness, irritability, hallucinations, and psychosis have occurred
Oxcarbazepine	Similar to carbamazepine, but hyponatremia is more common; fewer drug interactions
Phenobarbital	Sedation, cognitive dysfunction, tolerance, dependence, induction of hepatic drug metabolism; primidone is similar
Phenytoin	Nystagmus, diplopia, sedation, gingival hyperplasia, hirsutism, anemias, peripheral neuropathy, SJS/TEN rarely; induction of hepatic drug metabolism

Summary of Adverse Effects

Antiepileptic Drug	Adverse Effects
Tiagabine	Abdominal pain, nausea, dizziness, tremor, asthenia; drug metabolism is not induced
Topiramate	Drowsiness, dizziness, ataxia, psychomotor slowing and memory impairment; paresthesias, weight loss, acute myopia
Valproic acid	Drowsiness, nausea, tremor, hair loss, weight gain, hepatotoxicity (infants), inhibition of drug metabolism
Vigabatrin	Sedation, dizziness, weight gain; visual field defects/blindness with long-term use, which may not be reversible
Zonisamide	Dizziness, confusion, agitation, diarrhea, weight loss, rash, Stevens-Johnson syndrome

Questions

Thinking of valproate's mechanisms, therapeutic uses, and side effects, what are three takeaways regarding valproate?

- Valproate has multiple mechanisms of action.
- Valproate is a broad-spectrum ASD and first-line therapy in a range of seizure types in adults and children and has non-seizure applications (acute mania in biopolar disorder; migraine).
- Valproate has numerous and common GI and CNS side effects, tremor, hair loss, and weight gain. It can cause decreased platelet function, hyperammonemia, hepatotoxicity, and hypersensitivity reactions.

Thinking of topiramate's mechanisms, therapeutic uses, and side effects, what are three takeaways regarding topiramate?

- Topiramate has multiple mechanisms of action.
- Topiramate is a broad-spectrum ASD and used in a range of seizure types for adults and children. Topiramate has several non-seizure uses (weight loss, migraine prevention, and more).
- Topiramate has numerous and common CNS side effects. As a weak carbonic anhydrase inhibitor, it can cause decreased sweating and hyperthermia in children, reduced growth in children, and metabolic acidosis. Kidney stones may occur.

What about usage in pregnancy or individuals who may become pregnant?

- AVOID. Valproate is teratogenic. Topiramate may cause fetal harm.

Questions

A particular broad-spectrum ASD that inhibits NMDA and potentiates GABA at GABA_A receptors is used only in refractory seizures, such as Lennox-Gastaut syndrome. Informed consent is required. Why?

- Felbamate is associated with a high rates of fatal aplastic anemia and hepatic failure compared with patients taking other ASDs.

A particular class of drugs that acts at a presynaptic site is used for neuropathic pain, a number of other nonseizure uses, and as add-on therapy for management of focal seizures. What is its mechanistic target?

- Gapapentinoids block the $\alpha 2\delta$ subunit on VG Ca²⁺ channel on presynaptic neuron, which may decrease synaptic release of excitatory neurotransmitters.

A broad spectrum ASD with 100% bioavailability, nonhepatic metabolism, renal excretion of unchanged drug has mild-moderate side effects and low potential for drug interactions. What are the drug and the mechanistic target?

- Levetiracetam blocks SV2A, which may prevent docking of the vesicle and release of the neurotransmitter.

What are the most common side effects of the narrow spectrum drug for absence seizures?

- Nausea and vomiting, somnolence, sleep disturbance, hyperactivity (CYP3A4 DDIs)

What is the mechanism of action of GABA_A receptor positive allosteric modulators?

- The drugs bind an allosteric site on the GABA_A receptor and enhance the receptor's inhibitory action by increasing the frequency of channel opening → ↑ chloride ion influx.

What drugs fall into this class?

- Benzodiazepines and barbiturates.

What are the first-line drugs for the treatment of status epilepticus?

- Lorazepam, diazepam, and midazolam

When treating status epilepticus, what drug-drug combination increases the risk of respiratory depression?

- Benzodiazepine with phenobarbital

What is the explanation for the short duration of a single bolus dose of diazepam?

- Termination of effect by redistribution out of CNS tissue into muscle and fat tissues.

What are the antiseizure indications for pharmaceutical cannabidiol?

- Seizures associated with Dravet syndrome, Lennox-Gestaut syndrome, or tuberous sclerosis complex in patients ≥1 year of age